Physical Consequence, Not Localization, Governs Terrain-Induced Target Re-observability

First Author, Second Author, Third Author
Department, Organization
City, Country
first.author@example.edu

Abstract—A robot that has already observed a target can lose sight of it because its own motion and the intervening terrain block the line of sight. Restoring visibility is not a search problem: the robot knows approximately where the target is, and that knowledge is still not enough. This paper asks what information and what action reasoning terrain-induced target re-observability actually requires, and answers by elimination. Paired worlds that share terrain, pose, heading and a pixel-identical current observation still require opposite re-observation actions, with disjoint successful action sets in eight of eight pairs, so no memoryless rule can solve both; a second family additionally shares the target memory and still requires different optimal actions. Over 240 (state, action) pairs, 43.1% of the actions a geometric visibility oracle endorses fail when executed, while the same oracle placed at the pose the robot actually reaches agrees with the physical outcome 97.9% of the time, so the missing quantity is the physical consequence of an action rather than its visibility. On 2240 physically executed (state, action) pairs, adding the candidate action to observation plus memory raises re-observation AUC from 0.854 to 0.934, whereas adding the current observation to the memory buys 0.002. A planner built on the resulting action-conditioned consequence model reaches RSR@10 of 0.792 without privileged information, against 0.625 for uninformed random choice and 0.250 for every classical rule, while using 36 percent less motion than random choice. Three negative results bound how such a model may be used. Anchor recovery under perfect localization still fails physically in six of twenty-four terrains. A safety consequence head that is accurate and made better calibrated exactly where the decision threshold acts yields zero planning gain. Lookahead of two, three and five steps is not significantly better than one step on any budget while costing 1.8 to 2.1 times the motion. All results are in simulation with a vendored third-party locomotion policy; we claim the formulation, the benchmark, the failure analysis and the identification of the missing quantity, not a world model.

Index Terms—target re-observability, terrain-induced occlusion, action-conditioned consequence prediction, partial observability, legged locomotion, benchmark, failure analysis

I. Introduction

Robots operating in natural terrain rarely enjoy a clear view of what they care about. A legged platform descending into a crater, a rover traversing a ridge line, or a manipulator reaching behind a boulder all face the same structural problem: the terrain determines what can be seen from where, and the robot's own motion is what changes it. Partial observability here is not sensor noise but a deterministic consequence of geometry, and therefore predictable in principle — but only if the robot reasons about how its next action will change what it can see.

This paper studies a specific instance of that problem. The robot has already observed a target. Later the target is no longer visible, not because it moved or the sensor failed, but because the robot's own position and the intervening terrain now block the line of sight. The robot must choose an action that makes the target visible again. We call this target re-observability, and it differs from classical search in one decisive way: the robot has information about where the target is, and that information is still not enough.

The natural intuition that moving toward a target makes it more likely to be seen fails under terrain. Measured along a traverse toward a target beyond a ridge, the visibility sequence is not a single transition from visible to hidden but alternates:

terrain	profile	terrain	profile
stairs_up	V X V X	waves	X V X V X
trench	V X V X	perlin	X V X V X
pit	V X V X	basin	X V X
barrier	V X V X	perlin_rough	X V X V X

where V marks a position from which the target is visible and X one from which it is blocked. The number of transitions ranges from two to four on a single traverse. A policy that assumes monotonicity is not merely imprecise; it is aimed at the wrong objective.

Prior work supplies necessary components but addresses different questions. Lost-target search maintains a belief over where the target went and selects search actions to reduce it [1], [2]. Occlusion-aware active perception and next-best-view select viewpoints by predicting visibility or information gain from a candidate pose, typically using geometry rather than execution [3], [4]. Object permanence filters maintain a belief through periods of non-observation [5]. Embodied visual tracking and target-loss recovery is an established direction with its own benchmarks and systems [6], [7], and work in that line already combines action-conditioned world models with model-predictive control for recovering a lost target [8], while recent work couples visibility with reachability and feasibility [9]. None of this is unstudied, and we do not claim otherwise. What is missing is a unified analysis:

these lines are evaluated in their own settings, with their own observation and action interfaces and their own budgets, so the boundaries between them are not visible. We place the same decision rules on one task — terrain fixed, target static, action set identical, only the decision rule differing — and ask where and why each breaks.

This paper makes the following contributions. First, a formulation of terrain-induced target re-observability as an action-selection problem, with the non-monotonicity of visibility along a traverse as its defining property. Second, a procedurally generated benchmark of eight terrain-induced occlusion mechanisms, organised by physical mechanism and real-world analogue rather than as a difficulty ladder, with paired worlds that are provably identical in the current observation. Third, a frozen observation and action interface under which every decision rule is evaluated on the same terrain, budget and termination set, with all comparisons reported as paired bootstrap intervals. Fourth, a measured identification of the missing quantity, together with three negative results showing that accurate prediction is not the same thing as decision usefulness.

II. Related Work

Lost-target search maintains a belief over target location and plans search actions to reduce uncertainty [1], [2]. It answers which search action best reduces uncertainty, but not what to do when terrain makes the correct re-observation action point away from the belief. In our paired worlds the belief is correct and the action that restores visibility lies on the opposite side.

Occlusion-aware active perception and next-best-view score candidate viewpoints by predicted visibility or information gain [3], [4]. Candidate poses are typically evaluated kinematically, which assumes that an action takes the robot exactly where its nominal displacement says. We measure that this assumption fails, and that the visibility model itself is not the problem.

Object permanence maintains a belief through non-observation and acts on the filtered estimate [5]. It answers where the target probably is; it does not establish that knowing where the target is determines what to do. Our second pair family isolates exactly that gap.

Embodied visual tracking and target-loss recovery is an established direction [6], [7], with work already combining action-conditioned world models and model-predictive control for target recovery [8], and recent work coupling visibility with reachability and feasibility [9]. Our focus is narrower and physical: which quantity must be predicted for a known target to be made visible again under terrain-induced occlusion, and how that prediction should enter a decision.

III. Problem Formulation

Let O_t be the observation at time t, comprising pose, proprioception and a short action history; M_t the target memory, that is where and when the target was last seen; and $a_{t:t+H}$ a sequence of physical actions. Write $V_{t+h} \in \{0, 1\}$ for target visibility after executing that sequence. The task is

$$a^*_{t:t+H} \;=\; \arg\max_{a_{t:t+H}} \; P(V_{t+H} = 1 \mid O_t, M_t, a_{t:t+H}) \qquad (1)$$

The target is lost when $V_t = 0$ after having been observed, and re-observed when there exists $h \le H$ with $V_{t+h} = 1$. Physical safety is deliberately not part of this definition: Section VII shows that safety is a dimension of the consequence model rather than part of the task statement, and mixing it into (1) would confound the two.

The benchmark reports a difficulty curve rather than a single success flag, because the quantity of interest is how many physical actions are needed to regain sight. For episode i and budget K,

$$r_i(K) = \mathbf{1}[\exists\, h \le K : V_{i,t+h} = 1]\,, \qquad \mathrm{RSR@}K = \frac{1}{N}\sum_{i=1}^{N} r_i(K) \qquad (2)$$

Note what the definition does not require: visibility to be held. Visibility under these mechanisms is non-monotonic, and the task is to regain sight within a budget, not to keep it. The implementation is audited against (2) rather than assumed to match it: the metric is evaluated directly on the recorded per-step visibility trace, and a contract test asserts that this agrees with reading the termination state, so the headline number cannot drift if the episode semantics change.

Two companion metrics are reported and no single scalar combines them. The re-observation time T_{reobs} is the elapsed time from loss to the first re-observation. The action cost C_{action} is cumulative base-path length, averaged over every episode rather than only the successful ones so that exhausting the budget does not look cheap.

IV. Method

A. Visibility State and the Missing Quantity

H1, H2 and H3 below remove the simple explanations one at a time: the current observation does not determine the action, a correct target memory does not determine it either, and geometric visibility does not predict physical re-observation. The quantity that remains is the action-conditioned physical consequence. Rather than learning visibility end to end, we exploit the third result: since the visibility oracle is accurate at whatever pose the robot actually reaches, the model predicts the executed body-frame pose delta and the already-validated oracle is applied to the predicted pose. This keeps the validated part validated and localises the remaining error.

TABLE I
Terrain-induced occlusion mechanisms. A mechanism is a point on an axis, not a member of a category.

Mechanism	Axis	Real-world analogue
barrier	occlusion direction	walls, boulders, shelving
trench	elevation	ditches, trenches, ravines
stairs_up	elevation	staircases, platforms
pit	spatial scale	potholes, craters
basin	spatial scale	valleys, depressions
waves	visibility dynamics	dunes, rolling ground
perlin	terrain complexity	natural off-road ground
perlin_rough	terrain complexity	rubble, rock fields

B. Candidate Actions and the Consequence Model

Candidate actions are ten fixed locomotion primitives, each of the same simulated duration so that one step is one unit of budget. The consequence model is trained on 2240 physically executed (state, action) pairs, that is 24 terrains by 10 states by 10 actions, with positive rate 0.143. Features are the pose, the projected gravity vector, a short pose history, the target memory expressed in body coordinates with an explicit presence flag, and the action. Targets are the executed pose delta and whether the target was re-observed. Splits are by terrain and never random, so a terrain family cannot appear on both sides.

C. Closed-Loop Action Selection

The planner scores every candidate action with the model, takes the argmax of the predicted re-observation probability, and replans after each executed step, giving a receding-horizon controller over the learned model. An episode terminates at the first re-observation, at a fall, or on leaving the terrain.

V. Experimental Setup

A. Benchmark

The suite comprises eight terrain-induced occlusion mechanisms, listed in Table I and organised by physical mechanism rather than by surface name. Each is a situation in which a robot that has already seen a target loses sight of it and must regain sight by moving. The mechanisms span five roughly orthogonal axes: occlusion direction, elevation dependence, spatial scale, visibility transition frequency, and terrain spatial complexity. A mechanism is a point on an axis rather than a member of a category: pit and basin are both spatial scale, at opposite ends of it.

Two framing rules are enforced mechanically rather than left to intent. The taxonomy is checked for the wording challenging, hard, difficult or adversarial terrain and refuses to render if it appears, because the difficulty is in the target's visibility and not in the driving. And perlin and perlin_rough are not claimed to be real surfaces: Perlin noise is used as a procedural representation of spatially correlated natural elevation variation. All terrain is built as explicit box slabs rather than as a height field, so that the analytic visibility oracle and the collision surface are the same object.

B. Baselines and Metrics

Every baseline implements the core decision mechanism of a line of work rather than a full reproduction. Reactive always drives forward. HeadingLock holds the heading recorded at the loss moment. HistoryOnly aims at the remembered target. LostTargetSearch faces the belief and closes on it. ObjectPermanence runs a constant-velocity belief filter. NBV re-projects the believed target from a kinematic pose and is given the exact height field, since an active-perception method is allowed a map. Random draws uniformly from the frozen action set and establishes the lower bound. Oracle physically rolls out every candidate and is a privileged upper bound, never a deployable method.

Metrics are RSR@K as defined in (2), T_{reobs}, and C_{action}. Every proportion carries a Wilson score interval rather than a normal approximation, because several results sit at or near zero or one. Every comparison is paired: since all policies run on the same terrains, a difference in RSR is tested by resampling terrains with replacement and examining the distribution of the paired difference.

VI. Results

A. The Current Observation Does Not Determine the Action

We construct paired worlds that share terrain, robot pose, heading and a pixel-identical current observation, but in which the target was last seen from opposite sides. Across eight pairs the rendered camera images are identical with mean and maximum absolute pixel difference both zero, the optimal re-observation action is opposite in the two worlds in eight of eight, and the successful action sets are disjoint in eight of eight. Because any function of the current observation takes the same value in both worlds, no memoryless policy can solve both. On sixteen paired-history worlds the five classical rules score zero of sixteen.

B. A Correct Target Memory Is Also Not Sufficient

A natural reply is to remember where the target is. We therefore construct a second family of pairs sharing the current observation and the reconstructed target memory, differing only in terrain hidden inside the shadow of a visible occluder, which is invisible from the decision point and was never traversed. Over six such pairs the rendered observations are pixel-identical in six of six, the target memory is identical in six of six, and the optimal action differs in four of six. Two properties bound the claim. The successful action sets overlap, so this does not show that no fixed rule can succeed; it shows that a correct memory does not determine the optimal action. And on these pairs

TABLE II
Information ablation, leave-one-terrain-out, 2240 samples. Scores are ROC-AUC; pose error is mean absolute error over the four body-frame components. Best in bold.

Inputs	ROC-AUC	PR-AUC	Brier	ECE	Pose MAE (m)
O	0.759	0.360	0.117	0.076	0.436
M	0.852	0.639	0.077	0.039	0.461
$O{+}M$	0.854	0.600	0.084	0.056	0.425
$O{+}M{+}a$	0.934	0.748	0.062	0.022	0.148

TABLE III
Multi-step episode protocol, 24 terrains, budget 10×3.0 s. Every method uses the same action set, budget and termination set. Success is the re-observation rate; cost is mean base-path length over all episodes. Best non privileged result in bold.

Method	RSR@1	RSR@3	RSR@5	I
Random (lower bound)	0.083	0.542	0.625	
Reactive / HeadingLock / HistoryOnly	0.167	0.250	0.250	
NBV	0.167	0.208	0.250	
Anchor recovery (perfect)	0.167	0.250	0.250	
Consequence planner (ours)	0.708	0.708	0.792	
Oracle (privileged)	0.833	0.875	0.875	
Oracle search (privileged)	0.750	1.000	1.000	

all six classical policies choose the same action and fail in both worlds, so the construction is an information-level result and not a behavioural one.

C. Geometric Visibility Is Not Physical Re-observability

For every candidate action at every decision state we compute two quantities. $G(s,a)$ asks whether the target would be visible if the robot reached the ideal pose the action nominally produces, using a kinematic displacement and the line-of-sight oracle. $P(s,a)$ asks whether the target is actually re-observed when the primitive is executed. Over 240 pairs, the confusion matrix is $(G{=}0,P{=}0) = 171$, $(G{=}0,P{=}1) = 11$, $(G{=}1,P{=}0) = 25$ and $(G{=}1,P{=}1) = 33$. That is, 43.1% of the actions geometry endorses fail physically, and the disagreement is not one-sided: geometry also under-predicts in 4.6% of cases. Placing the same oracle at the pose the robot actually reaches makes it agree with the physical outcome in 97.9% of pairs. The pose prediction error is larger on the failing cell than on the agreeing cells by a paired bootstrap difference of +0.28 m with a 95% interval of $[+0.11, +0.45]$ m, decomposed into 0.52 m forward, 0.34 m lateral, 0.40 m in height and 5.5 degrees of yaw. No active-perception method is mis-implemented here; candidate viewpoints scored kinematically are simply a different quantity from where the robot ends up.

D. Information Ablation

Table II ablates the predictor input with terrain held out. Adding the candidate action raises AUC by +0.080, while adding the current observation to the memory raises it by +0.002. The information is in the action, not in seeing more. On the sixteen paired-world decision states the model reaches 0.81 accuracy and switches its chosen action in eight of eight pairs, which is what a memoryless rule provably cannot do.

E. Analytic Versus Learned Consequence

The learned pose model is compared against the analytic nominal displacement, which is what a kinematic predictor assumes. Mean absolute error falls from 0.469 to 0.148, a factor of 3.2, with per-component ratios of 5.5 for forward, 2.9 for lateral and 3.7 for yaw. The analytic model is nevertheless better for height, at 0.090 against 0.149, because it predicts no change and changes are small. The correct statement is that the learned model wins on the components that decide success, not that learning is better in all dimensions.

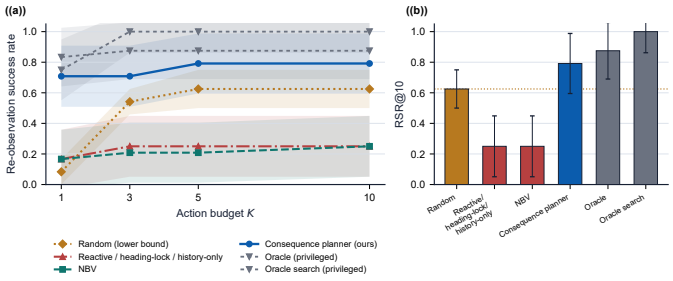


Fig. 1. Re-observation performance and the baseline ladder, over 24 terrains with a 10-step budget. (a) Success rate against action budget; intervals are Wilson 95% intervals, except for Random, which is a mean over three seeds and whose band is the spread across those seeds. (b) Success at the full budget with Wilson 95% intervals. Random, which never commits to a single action, sits above every classical rule; the consequence planner is higher still, at lower motion cost. The privileged oracles are reference bounds and are not deployable methods.

F. Consequence Prediction Converts Into Task Performance

Table III reports the multi-step protocol over 24 terrains with identical budget and termination set for every method. The consequence planner reaches RSR@10 = 0.792 and RSR@1 = 0.708 with no privileged information, against 0.625 for Random at 1.6 times the motion cost. By contrast, no classical baseline significantly outperforms any other: relative to HistoryOnly, Reactive, HeadingLock, LostTargetSearch and ObjectPermanence each differ by 0.000 with interval $[0, 0]$, that is, they succeed and fail on exactly the same terrain set. NBV differs by +0.042 with interval $[0.00, 0.125]$ and $p = 0.356$, so it cannot be claimed to be better. Only the privileged references separate.

G. Cross-Mechanism Generalisation

The unit of generalisation here is the mechanism, not the terrain. An earlier predictor held out a single terrain, which for the sixteen paired-history worlds still leaves

fifteen members of the same mechanism in training; that measures within-mechanism generalisation and cannot answer this question. We therefore hold out an entire mechanism family at once. The learned pose model beats the analytic baseline on all five held-out mechanisms, at a mean error of 0.329 against 0.491, so the causal structure from action to future physical state does transfer. Accuracy nevertheless degrades relative to the within-mechanism reference, where mean pose error is 0.196 and mean AUC 0.897 against 0.810 across mechanisms. The occlusion-direction fold reaches an AUC of 0.533, but that fold also removes 1680 of 2240 training rows, so it confounds the mechanism shift with a large reduction in data and cannot be attributed to mechanism transfer alone.

VII. Failure Analysis and Discussion

A. The Task Is Not Localisation Recovery

The most damaging reading of this benchmark is that it reduces to state recovery: save the pose at which the target was last seen and use odometry to return to it. We implement three such baselines — steering back to the last-visible pose, dead-reckoned backtracking along the recorded walk-in, and returning to that pose before advancing along the original sight ray — under perfect, moderate and realistic odometry drift. Perfect odometry does not solve the task: the best anchor-recovery baseline reaches RSR@10 = 0.250, the same as the classical rules and far below the consequence planner. More pointedly, six of its twenty-four failures under perfect localisation are physical terminations, meaning the robot falls or leaves the terrain while returning to a pose it knows exactly. Perfect and moderate drift produce identical results and realistic drift changes RSR@10 by 0.000, so the failure is not an odometry-accuracy problem but the physical consequence gap reached from an independent direction. Knowing exactly where the robot was does not imply that re-observation is physically executable.

B. Failure Type Changes With Planner Capability

Table IV classifies failures. Classical policies predominantly fail because they select ineffective actions, whereas the learned consequence planner substantially reduces action-selection failures and instead exposes physical termination failures. This shift is itself evidence that the modelled quantity is the right one and that the remaining gap concerns the coverage of the consequence model rather than candidate diversity. Our first version of this taxonomy recorded falls and boundary exits as action-selection failures, which is an attribution error: the action was chosen on a defensible prediction and what failed was the prediction of stability.

TABLE IV
Failure classification under protocol B, 24 terrains. B: belief present and a good action existed but was not chosen. F: the action was selected from a defensible prediction and the physical outcome was not anticipated.

Planner	B
HeadingLock / HistoryOnly / LostTargetSearch / ObjectPermanence	
Reactive	
NBV	
Consequence planner	
Oracle	

C. Better Prediction Is Not Better Decisions

Two negative results bound how a consequence model may be used, and both concern the prediction-to-decision step rather than prediction quality. Adding a safety consequence head raises the privileged oracle with ground truth from 0.3884 to 0.4107, so the term is worth having, and the learned head is accurate with ROC-AUC 0.900. Yet the planner gets worse at every penalty weight. Fixing the obvious explanation does not help: the head's overall calibration error is 0.042 while its error in the band where a threshold actually acts is 0.197, a factor of 4.7, and temperature scaling halves that to 0.099 and produces exactly zero planning improvement. Across raw, temperature-scaled and isotonic variants no operating point at any threshold or coverage beats not using safety at all.

A dependence analysis over the same 2240 samples explains why. Visibility and safety are negatively coupled, with an odds ratio of 0.35 and a 95% interval of $[0.249, 0.492]$, so the actions that reveal the target are less safe than those that do not, and the coupling persists within six of the ten individual actions, so it is not merely induced by action choice. But it almost never creates a decision conflict: safe actions are abundant at 9.17 per state and visible actions scarce at 1.43, and only five of 224 states have a visible action and a safe action but none that is both. The headroom for any joint decision is therefore bounded by that 2.2% conflict set, which is smaller than the noise in any learned head, and the joint head, a product of the two marginals, and a conformal filter all fail to improve on the visibility-only planner.

Longer lookahead fails in the same shape. Intermediate states in a chained rollout come from the learned model and never from ground-truth stepping, and only the first action of each plan is executed. Against a horizon of one step, paired success differences are -0.208 with interval $[-0.458, 0.042]$, 0.000 with $[-0.208, 0.208]$, and -0.125 with $[-0.333, 0.083]$ for horizons of two, three and five. No horizon is significantly better on any budget, and every horizon costs 1.78 to 2.09 times the motion. Lookahead does rescue two terrains that genuinely require multiple steps, which confirms that re-observation is intrinsically sequential, but it breaks eight terrains that one step solves. Model-error accumulation is not the explanation: the measured one-step pose error is lowest at a horizon of two and that horizon performs worst. The mechanism is

that a chained score taking a maximum over steps is monotonically optimistic in the horizon, so a longer lookahead offers more opportunities to find a sequence the model confidently believes will glimpse the target, and those are precisely the confidently wrong predictions. Longer rollout is not automatically beneficial when consequence uncertainty accumulates.

D. Limitations

All results are from simulation with a vendored third-party locomotion policy on a single controller, and no sim-to-real claim is made. The oracle is a privileged upper bound and the search reference is incomplete: an earlier version of this work judged four terrains unsolvable and beam search subsequently solved all four with two- to three-step sequences, so mistaking a planner's defect for a property of the benchmark nearly put a spurious ceiling into the results. The learned consequence model is evaluated with terrain held out, which is generalisation within a mechanism family. The target memory is a structured location representation reconstructed without noise, so belief-estimation failure is excluded by construction. The action set is ten fixed primitives and whether discretisation affects the paired-world conclusions has not been tested. A binary fall label proved unusable, with zero falls in 720 single-action rollouts, so stability is represented by boundary exit and a continuous margin, which means the learned safety head does not cover the event it is named after. Finally, the paired-world construction is available only in mechanisms that can express a directional aperture; five mechanisms where this is structurally impossible are documented in the benchmark definition.

VIII. Conclusion

We asked what information and what action reasoning a robot needs in order to regain sight of a target that terrain has occluded. The current observation is not sufficient, because worlds with pixel-identical observations can require opposite actions with disjoint successful sets. A correct target memory is not sufficient either, because worlds identical in observation and in memory can still require different actions. Geometric visibility is not sufficient, because 43.1% of the actions it endorses fail physically while the same oracle at the executed pose agrees with physics 97.9% of the time. What is missing is the action-conditioned physical consequence: adding the candidate action to observation plus memory raises predictive AUC by 0.080 while adding the observation to the memory raises it by 0.002, and a planner built on the resulting model reaches RSR@10 = 0.792 without privileged information, against 0.625 for uninformed random choice and 0.250 for every classical rule, while using 36% less motion than random choice.

Three negative results bound how such a model may be used, and all concern the prediction-to-decision step rather than prediction quality. Perfect localisation does not guarantee physical re-observability. A safety head that is accurate and made better calibrated at the decision boundary produces zero planning gain. Deeper lookahead is not significantly better on any budget, costs nearly twice the motion, and rescues two genuinely sequential terrains while breaking eight that one step solves. The compact statement of the paper is therefore that observation plus memory does not imply the optimal action, and the reason is the step in between: action to future physical state to future visibility must be modelled explicitly, must be action-conditioned, and must then be turned into a decision rather than a threshold.

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